

Program 1: Installing Maven and Gradle: Overview of Build Automation Tools, Key Differences Between Maven and Gradle, Installation and Setup

Java (JDK) Installation:

Step 1: Type the command **sudo apt install openjdk-11-jdk -y** in the terminal

It will install the OpenJDK 17 automatically.

```
samiksha@samiksha:~$ sudo apt install openjdk-11-jdk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-11-jdk is already the newest version (11.0.30+7-1ubuntu1~22.04).
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-tools-6.8.0-40 linux-image-6.8.0-40-generic
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic linux-tools-6.8.0-40-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 44 not upgraded.
samiksha@samiksha:~$
```

Step 2: Installation can be confirmed by running the command **java -version**

```
samiksha@samiksha:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-122.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-122.04.1, mixed mode, sharing)
samiksha@samiksha:~$
```

Maven Installation:

Step 1: Type the command **sudo apt install maven** in the terminal.

It will install Apache Maven from Ubuntu's package repository.

```
samiksha@samiksha:~$ sudo apt install maven
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
maven is already the newest version (3.6.3-5).
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-tools-6.8.0-40 linux-image-6.8.0-40-generic
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic linux-tools-6.8.0-40-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 44 not upgraded.
samiksha@samiksha:~$
```

Step 2: Installation can be confirmed by running the command **mvn -version**

```
samiksha@samiksha:~$ mvn --version
Apache Maven 3.6.3
Maven home: /usr/share/maven
Java version: 17.0.18, vendor: Ubuntu, runtime: /usr/lib/jvm/java-17-openjdk-amd64
Default locale: en_IN, platform encoding: UTF-8
OS name: "linux", version: "6.8.0-101-generic", arch: "amd64", family: "unix"
samiksha@samiksha:~$
```

Gradle Installation:

Step 1: Type the command **sudo apt install gradle** in the terminal

It installs Gradle build automation tool from Ubuntu's package repository.

```
samiksha@samiksha:~$ sudo apt install gradle
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gradle is already the newest version (4.4.1-13).
The following packages were automatically installed and are no longer required:
  linux-headers-6.8.0-40-generic linux-hwe-6.8-headers-6.8.0-40 linux-hwe-6.8-tools-6.8.0-40 linux-image-6.8.0-40-generic
  linux-modules-6.8.0-40-generic linux-modules-extra-6.8.0-40-generic linux-tools-6.8.0-40-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 44 not upgraded.
samiksha@samiksha:~$
```

Step 2: To check the proper installation of Gradle run the command **gradle --version**

```
samiksha@samiksha:~$ gradle --version

-----
Gradle 4.4.1
-----

Build time:   2012-12-21 00:00:00 UTC
Revision:     none

Groovy:       2.4.21
Ant:          Apache Ant(TM) version 1.10.12 compiled on January 17 1970
JVM:          17.0.18 (Ubuntu 17.0.18+8-Ubuntu-122.04.1)
OS:           Linux 6.8.0-101-generic amd64
```

